Validation of Right-Suite Canada Residential Load and Duct Calculations, Using HRAI's Windsor Example

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Validation of Right-Suite Canada Residential Load and Duct Calculations, Using HRAI's Windsor Example

Summary

This report compares load and duct calculations in Right-Suite Canada Residential (RSCR) to hand-calculated solutions from HRAI's Heat Loss/Heat Gain and Duct Design Manuals for a residential structure.

Using the Windsor Example from HRAI's Heat Loss/Heat Gain and Duct Design Manuals, it was found that the differences between the manual and computer generated load and duct size calculations were negligible, once the assumptions were identical in both methods. This report compares the calculations of RSCR to the manual calculations, explains how RSCR was used to design the heating and cooling system for the Windsor Example, and discusses differences in how the loads and ducts were calculated.

Introduction

The Windsor Example in HRAI's Heat Loss/Heat Gain and Duct Design Manuals (HRAI, 1996; 1998) is designed to be a detailed step-by-step example of F280 room-by-room load calculations and duct system design. This project is typical of Canadian residential construction (see Windsor house description in Appendix A). The house has thirteen rooms on three floors, including the basement. Heating and cooling equipment is located in the basement utility room, with all trunks also located in the basement. Risers in interior walls serve upper floors, and air is returned via stud wall plenums in interior walls. HRAI's manual calculation of the load is in Appendix B.

The Right-Suite Canada Residential (RSCR) program version 5.3.23 (Wrightsoft Corporation, 2001) is a fully integrated software system that generates a properly sized duct system, once the building and available equipment are entered into the program. Users enter the building description by drawing a floor plan with Wrightsoft's HVAC Shapes™. The load takeoff from the floor plan is automatic, and uses the required F280 method for load computation (CAN/CSA-F280-M90, 1990).

The most productive way to use Right-Suite is to preset commonly used parameters such as duct system type and fittings into a template, and to use the template as the starting point for a new project. In this case, a new template, *Canada Air AC – Gas* was used, available from the Canadian section of Wrightsoft's website, www.wrightsoft.com. A list of parameter settings in this template is shown in Appendix D.

To see step-by-step details of how to use RSCR to draw the Windsor Example, calculate the loads, and complete the ductwork, see Appendix F.

To better serve the building official community, a special Canada Inspection Report was developed, which includes specific details required for a building permit. The Inspection Report (Appendix K) combined with the duct plan printout (line drawings only; Appendix J) provides an acceptable package for inspectors. Full mechanical drawings showing detailed duct layouts may be required by some municipalities and will have to be prepared separately. It is recommended that contractors check with the local authority having jurisdiction as to their requirements.

Load Comparison

The initial result of the RSCR heating load calculations was compared with the manual solution (Appendix B). The resulting comparison (Figure 1) showed the total heating loads to be within 1,426 Btuh or 3.30%.

	Heating			
Room	**Manual**	**RSCR**	**Dif (Btuh)**	**Dif (%)**
Rec Rm	8,202	7,506	696	8.49
Utility	3,498	3,179	319	9.12
Front Ent	1,896	1,758	138	7.28
Liv Rm	5,255	5,020	235	4.47
Din Rm	3,604	3,393	211	5.85
Den	3,089	2,877	212	6.86
Kitchen	3,126	3,201	-75	-2.40
Entry Hall	1,899	1,899	0	0.00
Ensuite	1,939	1,968	-29	-1.50
Bath	1,346	1,575	-229	-17.01
Bdrm 3	2,820	2,813	7	0.25
Bdrm 2	3,049	3,003	46	1.51
Mstr Bdrm	3,471	3,576	-105	-3.03
Total	**43,194**	**41,768**	**1,426**	**3.30**

Figure 1. *Initial heating loads, manual and RSCR, as well as the difference in Btuh and the percent difference.*

One reason for the disparity between the loads was that the room areas in the two methods were different for the main and second floors. When comparing the two area calculations, the manual solution showed 62 ft^2 less overall area (out of 2550 ft^2). The area difference is due to the manual method's use of interior room dimensions while RSCR includes all interior walls (Appendix C). RSCR requires adjacent rooms to touch in order to determine which walls are exterior, and therefore contribute to heating and cooling loads. Appendix C shows how the actual dimensions used in the manual calculation leave out the partitions, and Appendix G shows the floor plan corrected for use in RSCR.

Leaving out the area of the interior partitions contributes a small error to the manual calculations. On the main floor, the omission of interior partitions in the manual method has the effect of eliminating a vertical strip of exterior wall on the outside end of each interior partition. On the second floor, in addition to the omission of the outside strip of wall area, there is a ceiling area omission along the top of each partition.

To be sure about the effects of these area differences, the manual calculation was entered as an Excel spreadsheet and the spreadsheet was modified to use the same areas as RSCR. With this approach, all differences due to dimensions were removed (Figure 2).

Note that in the RSCR building description, the header calculation was simplified by extending wall heights one foot throughout the basement and main floor.

	Heating			
Room	**Manual**	**RSCR**	**Dif (Btuh)**	**Dif (%)**
Rec Rm	8,194	8,198	-4	-0.05
Utility	3,496	3,496	0	0.00
Front Ent	1,920	1,877	43	2.24
Liv Rm	5,289	5,225	64	1.20
Din Rm	3,575	3,555	20	0.56
Den	2,917	3,050	-133	-4.57
Kitchen	3,181	3,133	48	1.51
Entry Hall	1,894	1,860	34	1.80
Ensuite	1,970	1,927	43	2.18
Bath	1,562	1,542	20	1.31
Bdrm 3	2,810	2,754	56	2.01
Bdrm 2	3,001	2,940	61	2.02
Mstr Bdrm	3,558	3,501	57	1.59
Total	**43,366**	**43,058**	**308**	**0.71**

Figure 2. *Heating loads after adjusting room area in the manual calculations*

The largest Btuh difference occurred in the Den. On careful inspection, it was noted that the differences in the Den were due to a different treatment of the header between the Den ceiling and the attic over the Kitchen. Since the Kitchen and Den are adjacent, the header between the two was correctly eliminated by RSCR. To make both cases have the same assumption, 14 ft^2 of exterior wall was added to the Den, simulating the exposed portion of the insulated header. See the Section View diagram in Appendix E.

Finally, the R-value used by RSCR differed from the R-value used in the manual calculations (11.41 versus 11.9); this difference was traced to the use of board sheathing, which the original project specified. A type 3B10 was used, while the manual calculations used a type 3A10 (types refer to RSCR categories of F280 materials). In the final recalculation of the manual method, the wall R-value was changed to 11.90 (Figure 3; Appendix H).

	Heating				Cooling			
Room	**Manual**	**RSCR**	**Dif (Btuh)**	**Dif (%)**	**Manual**	**RSCR**	**Dif (Btuh)**	**Dif (%)**
Rec Rm	8,233	8,198	35	0.43	400	376	-24	-6.31
Utility	3,513	3,496	17	0.47	168	155	-13	-8.24
Front Ent	1,874	1,877	-3	-0.17	149	156	7	4.34
Liv Rm	5,217	5,225	-8	-0.15	2,293	2,321	28	1.20
Din Rm	3,549	3,555	-6	-0.17	1,251	1,242	-9	-0.71
Den	3,045	3,050	-5	-0.17	926	920	-6	-0.64
Kitchen	3,128	3,133	-5	-0.15	2,633	2,616	-17	-0.64
Entry Hall	1,857	1,860	-3	-0.17	204	212	8	3.66
Ensuite	1,924	1,927	-3	-0.15	471	359	-112	-31.20
Bath	1,539	1,542	-3	-0.19	632	662	30	4.57
Bdrm 3	2,750	2,754	-4	-0.14	561	549	-12	-2.24
Bdrm 2	2,936	2,940	-4	-0.15	598	591	-7	-1.21
Mstr Bdrm	3,496	3,501	-5	-0.15	1,241	1,120	-121	-10.81
Total	**43,060**	**43,058**	**2**	**0.005**	**11,127**	**11,279**	**152**	**1.35**

Figure 3. *Final heating and cooling loads, with the additional 14 ft^2 of exposed header in the Den and wall R-values identical.*

Figure 3 shows a heating load difference of 2 Btuh, out of 43,058 Btuh, with a 0.005% difference overall. The remaining differences are so small that they cease to be of interest. Note that the maximum room-by-room variation is only 0.47%.

Sensible cooling loads are also close, with a 152 Btuh difference for an overall difference of 1.35%. It is important to remember that RSCR computes wall gains for cooling loads according to orientation, while the manual method uses no orientation information in calculating wall cooling loads.

Duct System Comparison

The template *Canada Air AC – Gas* includes typical sheet metal takeoffs and elbows in the duct preferences, sheet metal in the branches and trunks, and stud wall cavities in the return branches, as detailed in Appendix D. RSCR created a duct system based on the loads above.

The default duct design mode for RSCR calculates the duct system based on both the heating and cooling loads and chooses the larger of the ducts for the duct system. RSCR notes which ducts are heating or cooling based. Because the Windsor Example only considered heating loads, the fan flow rate was changed in RSCR to be 10 cfm for cooling and 810 cfm (per example, Appendix I) for heating. Reducing the fan flow rate to 10 cfm is equivalent to forcing RSCR to do a heating-only calculation for the duct system.

The heating-only duct system from RSCR was compared to the manually calculated duct system. Two branches out of 17 were different from the manual calculation. All trunk duct sizes differed; RSCR projected larger supply and smaller return trunks (Figure 4).

Supply Branch	Friction Loss		Velocity (fpm)		Duct Size (in.)	
	Manual	RSCR	Manual	RSCR	Manual	RSCR
S1	0.090	0.061	287	332	4	4
S2	0.090	0.055	413	413	4	4
S3	0.080	0.060	389	382	5	5
S4	0.090	0.067	367	360	5	5
S5	0.090	0.064	374	374	5	5
S6	0.090	0.062	346	341	6	6
S7	0.150	0.099	585	374	4	5
S8	0.120	0.087	367	360	5	5
S9	0.120	0.083	374	374	5	5
S10	0.110	0.077	413	401	4	4
S11	0.110	0.074	426	418	5	5
S12	0.080	0.055	418	404	5	5
S13	0.070	0.054	331	336	6	6
S14	0.130	0.092	344	332	4	4
S15	0.120	0.084	484	484	5	5
S16	0.130	0.093	413	401	4	4
S17	0.120	0.084	344	332	4	4
Supply Trunk	**Friction Loss**		**Velocity (fpm)**		**Duct Size (in.)**	
	Manual	**RSCR**	**Manual**	**RSCR**	**Manual**	**RSCR**
ST1 (A)	0.080	0.055	637	513	8 x 8	10 x 8
ST2 (B)	0.070	0.054	689	650	17 x 8	18 x 8
ST3 (C)	0.120	0.084	587	419	8 x 5	11 x 5

Return Branch	Friction Loss		Velocity (fpm)		Duct Size (in.)	
	Manual	RSCR	Manual	RSCR	Manual	RSCR
R1	0.060	0.086	377	337	14 x 3 ¼	14 x 3 1/4
R2	0.060	0.089	348	325	14 x 3 ¼	14 x 3 1/4
R3	0.040	0.054	366	354	2-14 x 3 ¼	2-14 x 3 1/4
R4	0.060	0.080	375	388	2-14 x 3 ¼	2-14 x 3 1/4
R5	0.070	0.099	425	416	12 x 3 ¼	14 x 3 1/4
Return Trunk	**Friction Loss**		**Velocity (fpm)**		**Duct Size (in.)**	
	Manual	**RSCR**	**Manual**	**RSCR**	**Manual**	**RSCR**
RT1 (D)	0.060	0.086	471	478	10 x 7	9 x 7
RT2 (E)	0.040	0.054	541	589	24 x 9	22 x 9

Figure 4. *First comparison of manual to RSCR heating only duct calculations. Shaded areas indicate a difference in duct dimensions between the manual and RSCR calculations.*

The difference in trunk sizes was attributed to the means of establishing the supply/return split for available pressure. The manual method uses a 50/50 supply/return pressure split for simple systems, while RSCR uses a split proportional to the effective length of the longest run in supply and return (40/60 supply/return split in this example). The HRAI Residential Air System Design Manual advises designers to use the proportional split for complex jobs (Note 2 on page 125; HRAI, 1998).

RSCR uses the proportional split for both simple and complex jobs to minimize velocities for the whole duct system, thereby creating a more efficient system.
A small contributor to the difference in trunk sizes is the difference in how the register and grille pressure drops are used in the calculations. While the manual method includes the register or grille as a separate pressure loss for each run, RSCR only allows one central estimate for supply registers and return grilles. In this case, RSCR allowed 0.02 for supply and 0.01 for return, while some supply registers in the manual calculation were 0.01 and some were 0.02.

The available pressure for each run is primarily influenced by supply/return pressure split. To be certain that the differences in the duct system design were due to the differences in supply/return pressure split, the pressure split used by RSCR was entered into the manual calculations. Again, an Excel spreadsheet was created for the manual calculations and the available pressure was modified. Figure 5 shows the resulting duct calculations.

Supply Branch	Friction Loss		Velocity (fpm)		Duct Size (in.)	
	Manual	RSCR	Manual	RSCR	Manual	RSCR
S1	0.060	0.061	332	332	4	4
S2	0.061	0.055	413	413	4	4
S3	0.058	0.060	381	382	5	5
S4	0.067	0.067	359	360	5	5
S5	0.064	0.064	374	374	5	5
S6	0.063	0.062	341	341	6	6
S7	0.104	0.099	374	374	5	5
S8	0.087	0.087	359	360	5	5
S9	0.083	0.083	374	374	5	5
S10	0.076	0.077	413	401	4	4
S11	0.075	0.074	425	418	5	5
S12	0.055	0.055	403	404	5	5
S13	0.053	0.054	336	336	6	6
S14	0.089	0.092	332	332	4	4
S15	0.083	0.084	484	484	5	5
S16	0.088	0.093	401	401	4	4
S17	0.082	0.084	332	332	4	4
Supply Trunk	**Friction Loss**		**Velocity (fpm)**		**Duct Size (in.)**	
	Manual	**RSCR**	**Manual**	**RSCR**	**Manual**	**RSCR**
ST1 (A)	0.058	0.055	568	513	9 x 8	10 x 8
ST2 (B)	0.053	0.054	650	650	18 x 8	18 x 8
ST3 (C)	0.082	0.084	548	419	10 x 5	10 x 5

Return Branch	Friction Loss		Velocity (fpm)		Duct Size (in.)	
	Manual	RSCR	Manual	RSCR	Manual	RSCR
R1	0.080	0.086	339	337	14 x 3 1/4	14 x 3 1/4
R2	0.080	0.089	326	325	14 x 3 1/4	14 x 3 1/4
R3	0.052	0.054	353	354	2-14 x 3 1/4	2-14 x 3 1/4
R4	0.076	0.080	388	388	2-14 x 3 1/4	2-14 x 3 1/4
R5	0.093	0.099	415	416	14 x 3 1/4	14 x 3 1/4
Return Trunk	**Friction Loss**		**Velocity (fpm)**		**Duct Size (in.)**	
	Manual	**RSCR**	**Manual**	**RSCR**	**Manual**	**RSCR**
RT1 (D)	0.080	0.086	480	478	9 x 7	9 x 7
RT2 (E)	0.052	0.054	588	589	22 x 9	22 x 9

Figure 5. *Both the manual and RSCR calculations use the same pressure split for this figure (heating only). Shaded areas indicate a difference in duct dimensions between the manual and RSCR calculations.*

By changing the pressure split in the manual calculations to match RSCR, all but one of the duct sizes were identical. The one exception is Supply Trunk 1 (or 'A' in the manual calculation). RSCR recommends using a 10"x 8" duct, which is 1 inch wider than the one recommended by the modified manual calculations. To find out why this difference occurred, the round dimensions from both calculations were compared. Both calculations suggested a 10" duct size. The rectangular duct size equivalent table that the HRAI Manual uses (Table C6 pg.128; HRAI, 1998) yields a different

equivalent than RSCR. The HRAI table skips from 9”x 8” to 11”x 8” apparently due to rounded decimal places. Either duct size would be appropriate in practice. Note that the effects of central versus individual register and grille pressure drops appear to be very minor in this case.

The usual mode for RSCR is to design duct systems used for both heating and cooling. Since a heating-only duct system could be undersized for cooling, it is important to realize that the Windsor example may not be an adequate duct system for cooling loads. In order to design a system for both heating and cooling, it’s necessary to compute both loads, and then to compute the airflows required for both loads. As a final step in the design process, RSCR designs each run for the larger load and notes the dominant load with either an ‘h’ or a ‘c’ on the duct branch and trunk screens and in printed reports. We compared the RSCR duct system created for both heating and cooling (by resetting the cooling cfm to the default 810 cfm’s) to the RSCR system calculated for the Windsor Example (heating-only) (Figure 6).

Supply Branch	Duct Size (in.)		HTG or CLG Based
	HTG Only	HTG & CLG	
S1	4	5	**C**
S2	4	4	H
S3	5	5	H
S4	5	6	**C**
S5	5	5	H
S6	6	6	**C**
S7	5	5	H
S8	5	6	**C**
S9	5	5	H
S10	4	4	H
S11	5	5	**C**
S12	5	5	H
S13	6	6	**C**
S14	4	6	**C**
S15	5	5	H
S16	4	4	H
S17	4	6	**C**
Supply Trunk	**Duct Size (in.)**		**HTG or CLG Based**
	HTG Only	**HTG & CLG**	
ST1 (A)	10 x 8	10 x 8	**C**
ST2 (B)	18 x 8	18 x 8	H
ST3 (C)	11 x 5	14 x 5	**C**

Return Branch	Duct Size (in.)		HTG or CLG Based
	HTG Only	HTG & CLG	
r1	14 x 3 1/4	14 x 3 1/4	**C**
r2	14 x 3 1/4	14 x 3 1/4	H
r3	2-14 x 3 1/4	2-14 x 3 1/4	H
r4	2-14 x 3 1/4	2-14 x 3 1/4	**C**
r5	14 x 3 1/4	14 x 3 1/4	H
Return Trunk	**Duct Size (in.)**		**HTG or CLG Based**
	HTG Only	**HTG & CLG**	
RT1 (D)	9 x 7	9 x 7	H
RT2 (E)	22 x 9	22 x 9	**C**

Figure 6. *Comparison of duct sizes for a heating only duct system to a system designed for heating and cooling.*

The heating and cooling duct system had 13 cooling based ducts out of a possible 27 ducts. Of the 13 cooling based ducts, RSCR chose larger duct sizes for 6 of the ducts (5 supply branches and 1 supply trunk) because of the cooling loads for the rooms served by these ducts.

Discussion and Conclusion

From this detailed study, it is clear that Right-Suite Canada Residential yields essentially the same load and duct calculations as the manual calculations, if the assumptions are the same as in a manual calculation.

Two inconsistent assumptions, identified through close examination of the load calculations, were the treatment of room areas and insulated headers. The room areas used in the manual calculations are the interior dimensions; these dimensions are used to simplify the calculations for computation by hand. RSCR requires room dimensions that include interior walls for proper load calculation and for identification of interior room partitions. RSCR's ability to properly account for partitions and the entire interior floor area is a benefit of the computer application. Resulting loads are more accurate, with no extra effort.

There are two ways to account for the insulated headers in RSCR. One way is to add the header to the basement and main floor (excluding the kitchen and entrance hall) ceiling heights. Another method that could be used to include the insulated headers into the load calculations is to enter them into RSCR, via property sheets, as an additional wall type. Both yield similar results, though the former method requires less effort.

There were three assumptions associated with the RSCR duct system that conflicted with the manual calculations. RSCR calculates a duct system based on both heating and cooling loads, splits the fan pressure proportional to the equivalent length in the supply and return paths, and assumes that register pressure loss is the same in all branches.

By default, RSCR calculates a duct system using both the heating and cooling loads. To eliminate the cooling loads from the duct design calculations, the fan pressure was changed to a very low flow rate. This forces RSCR to calculate the duct system using only heating loads.

RSCR uses a pressure split proportional to the equivalent length in the supply and return paths. It is the preferred method because it produces the best system designs by minimizing velocities in the entire system. A 50/50 supply/return pressure split produces an acceptable design but it does not maximize efficiency in most cases. This split has been used by HRAI to simplify calculations by hand. In the Windsor Example, RSCR used approximately a 40/60 supply/return split. A 50/50 split decreases size in the supply and increases size in the return when compared to a 40/60 split. The result is higher velocities in the supply (increased noise) and lower velocities in the returns (decreased noise) with the manual method. Because the proportional split reduces both supply and return to a minimum overall velocity, it also provides the quietest overall system.

Manual calculations determine pressure losses for each branch, allowing individual branches to have different pressure losses for each register. This is only necessary when the supply register pressure drop becomes significant compared to the pressure drop in the supply branch path. Given typical register pressures of 0.01 to 0.03 inches of water and a typical supply side pressure of 0.15, there is very little likelihood of any difference in branch size.

Overall, RSCR generates accurate load calculations and duct system designs compliant with F280. Furthermore, RSCR produces more precise calculations by using more advanced methods that are difficult and time consuming to compute by hand.

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